Coen Hao

coen.hao@mail.utoronto.ca | linkedin.com/in/coen-hao-04014b209 | refriderator.github.io/Portfolio | Toronto, ON

Physics undergraduate with 5+ years of hands-on EFI tuning, suspension design, and vehicle-dynamics work across Formula Student and international motorsport competition.

EDUCATION

University of Toronto *Expected 2028*

Honours B.Sc. — Physics Major; Mathematics Minor; Statistics Minor

Relevant coursework: Classical Mechanics, Practical Physics (Python), ODEs, Electricity & Magnetism, Linear Algebra, Probability, Statistics & Data Analysis

EXPERIENCE

Suspension & Vehicle Dynamics — UTFR Formula Student *2024 – 2026*

- Compared suspension setups against logged data using MATLAB to model lap-time effects.
- Fabricated uprights and performed alignment and ride-height adjustments before and during competition.
- Wrote driver-feedback reports and conducted iterative on-track testing.

Driver & Competition Vehicle Build — Team Opposite Lock *2021 – Present*

- Designed, budgeted, and built a turbo system combining OEM engine layouts, including oil/coolant feed and return lines and data logging. Validated the build with a 1,500-mile road trip and 8+ hours of competition driving.
- Tuned a Haltech standalone ECU with closed-loop boost, air-fuel, and PID idle control. Isolated idle faults across mechanical, electrical, and software causes.
- Increased steering lock 20° by relocating the tie-rod mounting location on the knuckle. Corrected the resulting Ackermann and bump steer characteristics by building 2 cm steering rack spacers specced from basic geometry simulations.
- Built and modified car/tire physics models and Assetto Corsa 3D assets from real-world scans.
- Competition driver at Final Bout Special Stage Central 2025 (international team-tandem drifting) against teams from Japan, Australia, Europe, and South America.

EFI Tuning & Powertrain — Independent / Competition Vehicles *2021 – Present*

- Tuned 5 competition vehicles (forced-induction, ITB, and rotary setups) on street and dyno, winning across multiple disciplines of racing, including 3 first-place vehicles in MotorTrend's Roadkill Nights autocross.
- Rebuilt 3 full engine and body harnesses using OEM documents, oscilloscope testing, repinning, and hand soldering.
- Modeled and optimized intake and exhaust airflow using CFD and 3D-printed prototyping.

TECHNICAL SKILLS

Simulation & Analysis: MATLAB, Simulink, Adams Car, CFD, Python, suspension geometry & kinematics, data logging

Powertrain & Electrical: Dyno/EFI tuning, PID tuning, oscilloscope diagnosis, wiring harness fabrication, Arduino C

Fabrication: Fusion 360, upright fabrication / jigging, alignment & corner-weighting, fiberglass repair & forming

Other: Sponsorship & professional relations, budgeting & financials, event logistics